Lecture notes

Recap.

$$D = \{(x^s, y^s)\}_{s=1}^{S}, \quad x^s, y^s \in \mathbb{R}^n$$

Encoder-decoder architecture

All hidden layer & output layer neurons have **linear** activations.

For input x, the prediction is:

$$\hat{y} = ABx$$

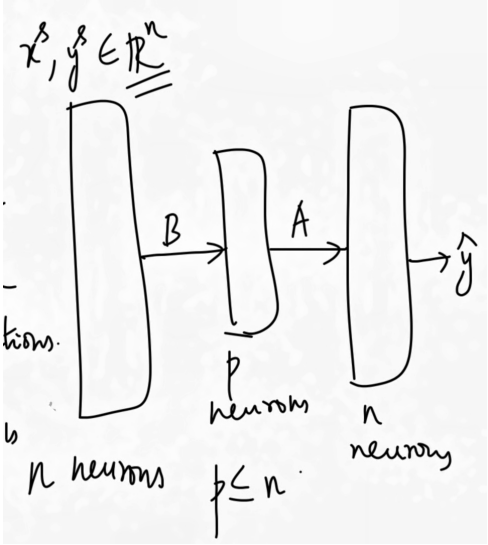


Figure 1: A chain of three boxes — p-neuron layer $\xrightarrow{B}$ n-neuron layer ($p \leq n$) $\xrightarrow{A}$ output layer $\xrightarrow{\hat{y}}$.

Loss fn.: Squared error.

For $D = \{(x^s, y^s)\}_{s=1}^{S}$,

$$E(A, B) = \sum_{s=1}^{S} \|y^s - y^*\|^2 = \sum_{s=1}^{S} \|ABx^s - y^s\|^2$$

$$\xrightarrow{b}\xrightarrow{a}\rightarrow$$

$$\sum_{s=1}^{S}(abx^s - y^s)^2 =: E(a,b)$$

When a is fixed: observe that $E(a_{fixed}, b)$ is **convex** wrt b. When b is fixed, $E(a, b_{fixed})$ is cvx wrt a.

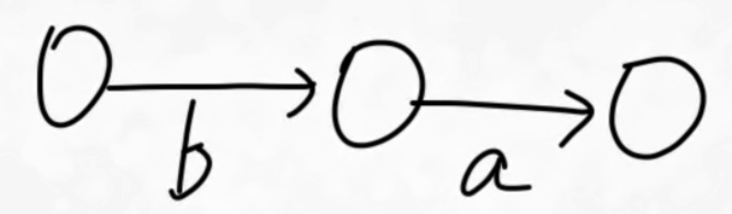


Figure 2: drawing from the page (no description in the notes)

Convexity:

Convex combination of 2 pts.

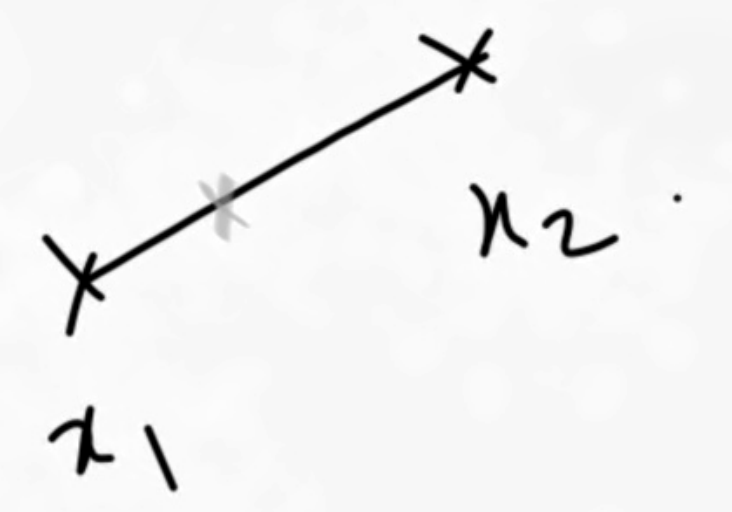


Figure 3: A line segment with two points x_1 (left) and x_2 (right), marked by an asterisk at the midpoint $\lambda x_1 + (1-\lambda)x_2$, `\lambda \in \[0,1\]` (not typeset).

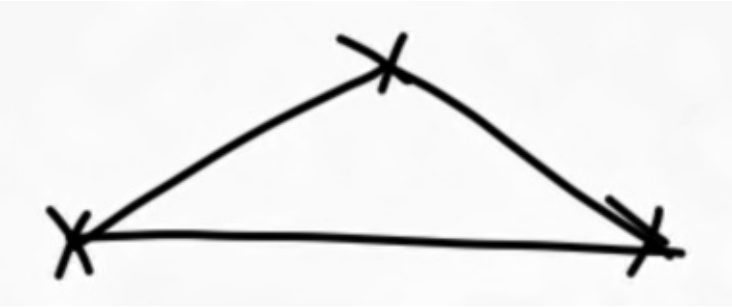


Figure 4: A triangle with a convex combination point marked inside it.

A set C is convex if for any arbitrary pair $x_1, x_2 \in C$, the cvx combination $\lambda x_1 + (1-\lambda)x_2$ also belongs to C.

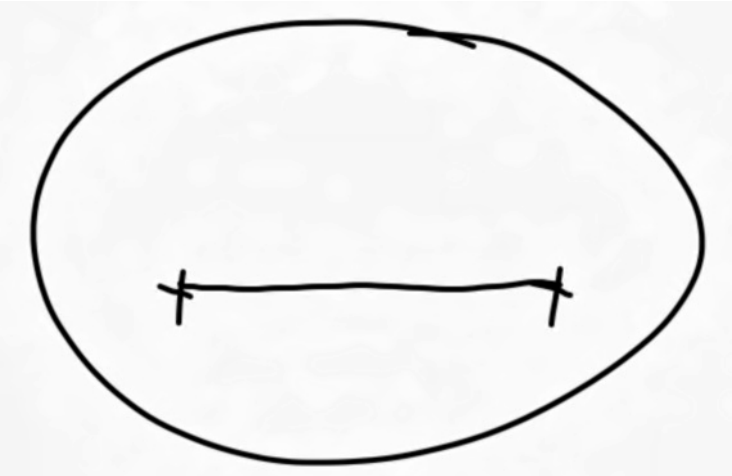


Figure 5: An oval shape with a horizontal line segment drawn through its center.

Convex function:

A function $f : C \to \mathbb{R}$ over a convex set C is called **convex** if:

for any $x_1, x_2 \in C$ and for $\lambda \in [0, 1]$

$$f(\lambda x_1 + (1-\lambda)x_2) \leq \lambda f(x_1) + (1-\lambda) f(x_2)$$

cvx comb of x_1 & x_2

Convex combination of fn evaluated at x_1, x_2

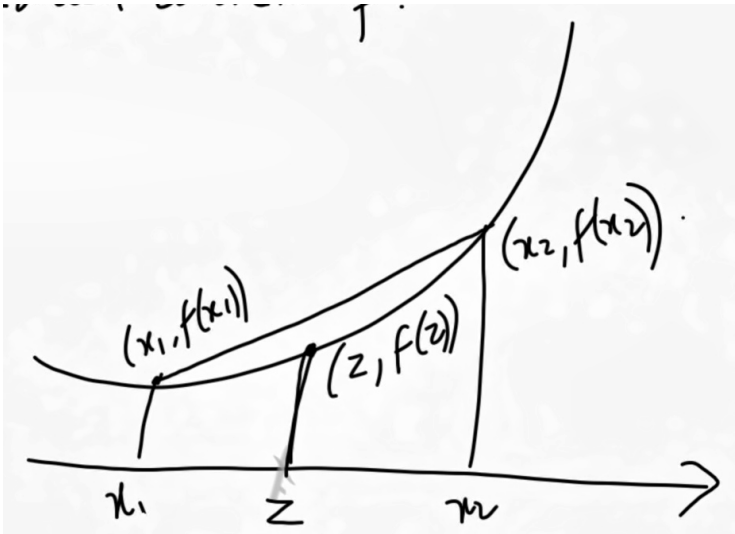


Figure 6: A curve with points $(x_1, f(x_1))$, $(z, f(z))$, $(x_2, f(x_2))$ on it, showing the chord between $(x_1, f(x_1))$ and $(x_2, f(x_2))$ lying above the curve at point z.

graph $\leq$ Chord

Optimization problem:

$$\min_{x \in C} f(x) \rightarrow \text{(P)}$$

convex set $\rightarrow$ convex fn.

(P) is a convex opt problem

$$\min_{w_1, w_2, \ldots, w_L} E(\hat{y}, y) \quad \hat{y} = \text{MLP}(x; w^1, w^2, \ldots, w^L)$$

Characterizations of convex fn:

1) If f is cont. diff. then it satisfies some cdn. [Diagram: A bowl-shaped curve with a tangent line at its minimum point.]
2) If f is twice cont. diff. then: $f''(x) \geq 0 \; \forall x \in C$

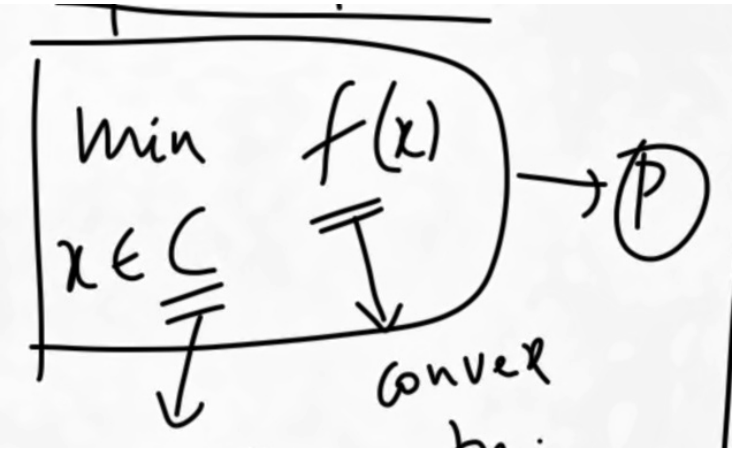


Figure 7: drawing from the page (no description in the notes)

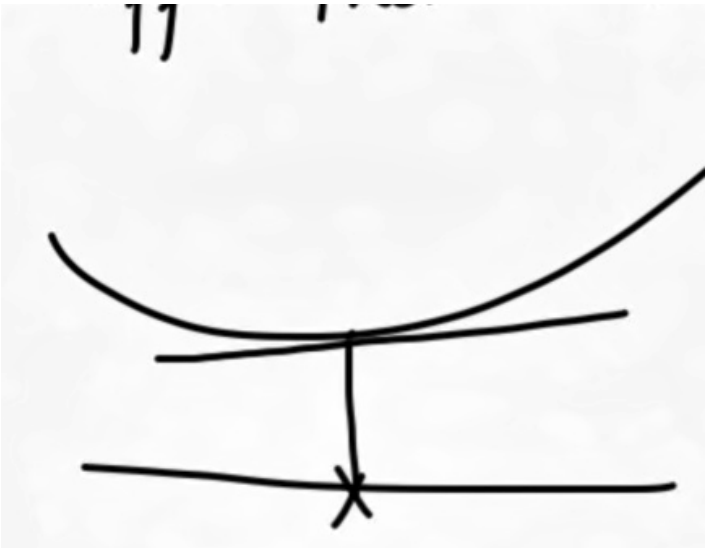


Figure 8: drawing from the page (no description in the notes)

In n-dim case:

1) $\nabla_x^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$

2) for twice diff f, $\nabla_x^2 f \geq 0$

Hessian if p.s.d. $\rightarrow$ the eigen values of $\nabla_x^2 f$ are non-negative.

A matrix A is positive semi-def if:

$$v^T A v \geq 0 \quad \forall v \in \mathbb{R}^n$$

$$\langle v, Av \rangle \geq 0$$

$$\min_{A,B} \sum_{s=1}^{S} \|ABx^s - y^s\|_2^2 =: E(A,B) \quad x^s \in \mathbb{R}^n,\ y^s \in \mathbb{R}^n$$

When wts of A are fixed, is $E(A_{fixed}, B)$ convex wrt wts of B? When wts of B are fixed, is $E(A, B_{fixed})$ convex wrt wts of A?

$$\sum_{s=1}^{S} \|\hat{y}^s - y^s\|_2^2$$

$$Y = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ y^1 & y^2 & \cdots & y^S \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

sample 1 true, sample 2 true, ..., sample S true

$$\hat{y} = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \hat{y}^1 & \hat{y}^2 & \cdots & \hat{y}^S \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}$$

$$\sum_{s=1}^{S} \|\hat{y}^s - y^s\|^2 =$$

$$-\left\| \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ \hat{y}^1 & \hat{y}^2 & \cdots & \hat{y}^S \\ \downarrow & \downarrow & & \downarrow \end{bmatrix} \right\|^2 =$$

$$\|\text{vec}(\hat{Y}) - \text{vec}(Y)\|^2 = E(A,B)$$

vec(Y) = vectorization/flattening of Y

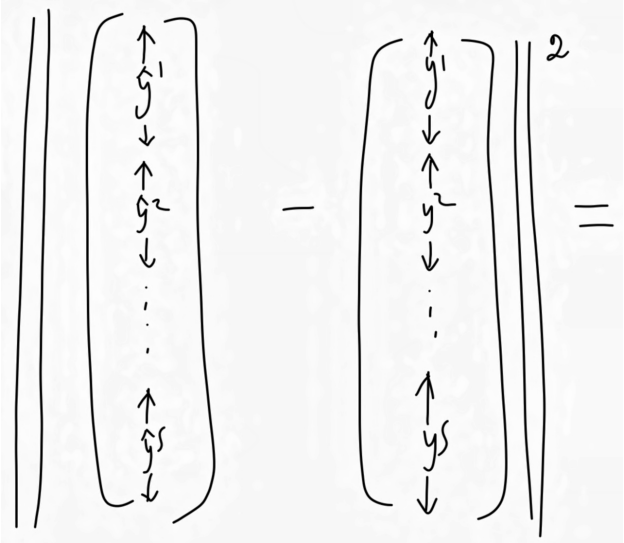


Figure 9: A column of $\hat{y}^1, \hat{y}^2, \ldots, \hat{y}^S$ (each with an up arrow above and down arrow below) next to a column of $y^1, y^2, \ldots, y^S$ (each with an up arrow above and down arrow below), separated by a vertical line.

$$= \begin{bmatrix} \uparrow \\ y^1 \\ \downarrow \\ \uparrow \\ y^2 \\ \downarrow \\ \vdots \\ \uparrow \\ y^S \\ \downarrow \end{bmatrix}$$

$$\hat{y}^1 = ABx^1, \quad \hat{y}^2 = ABx^2, \quad \vdots \quad \hat{y}^S = ABx^S$$

$$E(A, B) = \|\text{vec}(ABX) - \text{vec}(Y)\|^2$$

$$AB \begin{bmatrix} \uparrow \\ x^1 \\ \downarrow \\ \uparrow \\ x^2 \\ \downarrow \\ \vdots \\ \uparrow \\ x^S \\ \downarrow \end{bmatrix} = \text{vec}(ABX)$$

$$E = \|\text{vec}(ABX) - \text{vec}(Y)\|^2$$

When we fix A, $E = \|M\text{vec}(B) - \text{vec}(Y)\|^2$

Suppose $Z = \text{vec}(B)$, $C = \text{vec}(Y)$

$$E = \|MZ - C\|_2^2 = (MZ - C)^T(MZ - C) = (Z^T M^T - C^T)(MZ - C) = Z^T M^T MZ - 2(MZ)^T C + C^T C$$

$$\min_Z \|MZ - C\|^2 / \text{convex opt problem}$$

Opt. problem is convex wrt. Z

$$F(Z) = \|MZ - C\|^2, \quad \nabla F(Z) = 2M^T MZ - 2M^T C$$
$$\nabla^2 F(Z) = 2M^T M, \quad M^T M \text{ is p.s.d.}$$

M is of some size $q \times q$, for any $v \in \mathbb{R}^q$, $v^T M^T Mv \geq 0$. $\|Mv\|^2 \geq 0$



Figure 10: A convex curve with a minimum marked at its lowest point.

$$\nabla F(z) = 2M^T Mz - 2M^T C = 0$$

If z^* satisfies $M^T Mz^* - M^T C = 0$, then z^* is optimal pt. for min $\|Mz - c\|$.

Optimality for cvx opt. problems: A pt. $x^* \in C$ is an optimal point for the cvx. opt. problem $\min_{x \in C} f(x)$ if and only if $\nabla f(x^*) = 0$

$$\|\text{vec}(ABX) - \text{vec}(Y)\|^2 = \epsilon$$

for fixed A, $\|M\text{vec}(B) - \text{vec}(Y)\|^2$

$$\|Mz - c\|_2$$

$$\text{vec}(ABX) = (X^T \otimes A)\text{vec}(B)$$

Kronecker product of X^T & A

Suppose $G = \begin{bmatrix} g_{11} & g_{12} & \cdots & g_{1n} \\ \vdots & & & \vdots \\ g_{m1} & g_{m2} & \cdots & g_{mn} \end{bmatrix}_{m \times n}$

$$H = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1q} \\ \vdots & & & \vdots \\ h_{p1} & h_{p2} & \cdots & h_{pq} \end{bmatrix}_{p \times q}$$

$$G \otimes H = \begin{bmatrix} g_{11}H & g_{12}H & \cdots & g_{1n}H \\ \vdots & & & \vdots \\ g_{m1}H & g_{m2}H & \cdots & g_{mn}H \end{bmatrix}_{mp \times nq}$$

$$\|Mz - c\|^2 =: F(z)$$
$$\|(X^T \otimes A)z - c\|^2$$
$$\nabla F(z) = 0 \Rightarrow M^T M z - M^T c = 0$$

For an opt-z^*,

$$M^T M z^* = M^T c$$
$$(X^T \otimes A)^T (X^T \otimes A) z^* = (X^T \otimes A)^T c$$

Prop 1: $(G \otimes H)^T = G^T \otimes H^T$

Prop 2: $(G \otimes H)(U \otimes V) = GU \otimes HV$

$$(X^T \otimes A)^T (X^T \otimes A) z^* = (X^T \otimes A)^T c$$
$$(X \otimes A^T)(X^T \otimes A) z^* = (X \otimes A^T) c$$
$$(XX^T \otimes A^T A) z^* = (X \otimes A^T) c$$

$$\text{vec}(ABX) = (X^T \otimes A)\text{vec}(B)$$

$$(XX^T \otimes A^T A)\text{vec}(B^*) = (X \otimes A^T)\text{vec}(Y)$$
$$A^T A \; B^* \; XX^T = A^T Y X^T$$
$$XX^T = \quad Y X^T$$

$$X = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ x^1 & x^2 & \cdots x^S \\ \downarrow & \downarrow & \downarrow \end{bmatrix}_{n \times S}$$

$$XX^T = \text{feature covariance matrix}, \quad n \times n \text{ matrix}$$

$$XX^T = \sum_{s=1}^{S} x^s (x^s)^T, \quad YY^T = \sum_{s=1}^{S} y^s (x^s)^T, \quad XY^T = \sum_{s=1}^{S} x^s (y^s)^T$$

$$A^T A B^* X X^T = A^T Y X^T$$

$$\boxed{A^T A B^* \Sigma_{xx} = A^T \Sigma_{yx}}$$

Assume: - Σ_{xx} is invertible, - A is full rank $\Rightarrow A^T A$ is also inv.

$$A^T A B^* \Sigma_{xx} = A^T \Sigma_{yx}$$

$$B^* = (A^T A)^{-1} A^T \Sigma_{yx} (\Sigma_{xx})^{-1}$$

$\rightarrow$ **Unique soln?**

$$\min_z \|Mz - q\|^2$$

$\sum_{xx}$ is invertible?

Linear comb. of $v_1, v_2 \dots v_k$: $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k$, $\alpha_1, \dots, \alpha_k$ are real nos.

A set $v_1, v_2 \dots v_k$ is l.i. if $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k = 0$ is written uniquely for $\alpha_1, \alpha_2 \dots \alpha_k$.

$X = \begin{bmatrix} \uparrow & \uparrow & \uparrow \\ \vdots & \vdots & \vdots \\ x^1 & x^2 & \dots x^S \end{bmatrix}$ has linearly independent columns.

$$v_j \neq \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{j-1} v_{j-1} + \beta_{j+1} v_{j+1} + \cdots + \beta_k v_k$$

A is full rank

[Diagram: A chain $B \to D \to A \to n$, with $p \leq n$.]

$$A_{n \times p} \text{ matrix}, \quad p \leq n$$

$$A_{n \times p}\, h_{p \times 1} \to b_{n \times 1} \text{ vector}$$

$$\text{rank}(A) = p \iff A^T A \text{ is positive definite}$$

$$\text{rank}(A) = \text{num of linearly ind rows/cols}$$

$$\text{rank}(A) \leq \min(p, n) \quad \text{(if } A^T A \text{ is invertible)}$$

$$\text{rank}(A) \leq p \quad \text{when } A \text{ is full rank, rank}(A) = p$$

$$\nabla^2 F = M^T M \text{ is p.d.} \implies \|Mz - c\|^2 \text{ becomes strictly convex}$$

$$\left(X^T \otimes A\right)^T \left(X^T \otimes A\right)$$
$$\left(X \otimes A^T\right) \left(X^T \otimes A\right)$$
$$\left(XX^T \otimes A^T A\right) \to \text{p.d.}$$

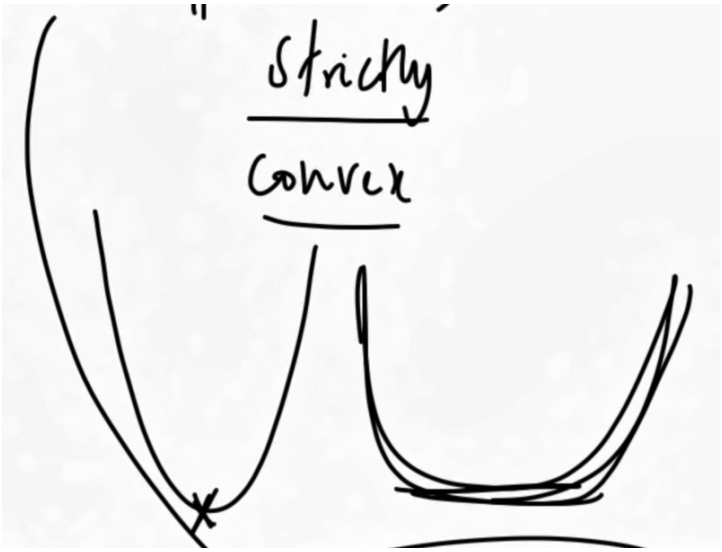


Figure 11: A strictly convex bowl shape with a unique minimum marked.

eigen values are strictly +ve

minimizer is unique

Prop 3: If G is sym.p.d & H is sym.p.d then $G \otimes H$ is also sym.p.d.

$$A = \begin{bmatrix} \uparrow & \uparrow & & \uparrow \\ A_1 & A_2 & \cdots & A_p \\ \downarrow & \downarrow & & \downarrow \end{bmatrix}, \quad h = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_p \end{bmatrix}$$

[Diagram: $D_B \xrightarrow{B} D_A \xrightarrow{A} D$, chain of three boxes.]

$$Ah = h_1 A_1 + h_2 A_2 + \cdots + h_p A_p \quad \text{(b's)}$$

$$\min_{A,B} \|\text{vec}(ABX) - \text{vec}(Y)\|$$

$$W = AB$$
$$E(A,B) = \text{Tr}(\Sigma_{yy}) - \sum_{i \in F} \lambda_i$$

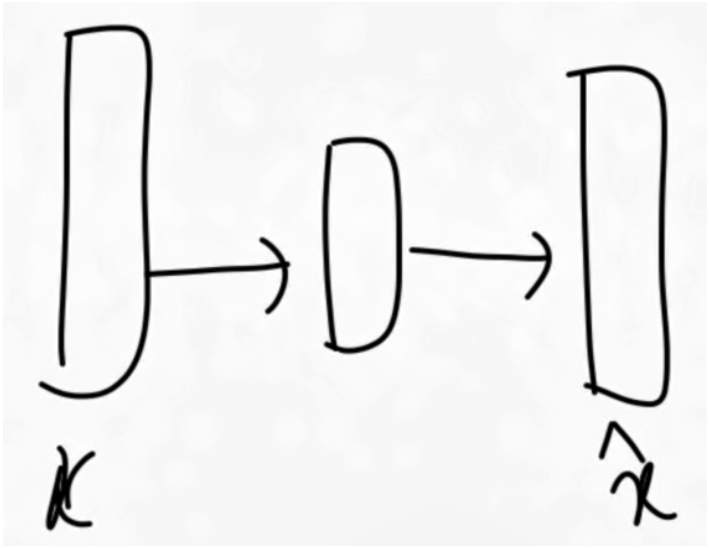


Figure 12: A chain $x \to B \to \hat{x}$

$$\Sigma = \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{xy} = \Sigma_{xx}\Sigma_{xx}^{-1}\Sigma_{xx}$$

$$\sum_s \|ABx^s - x\|^2, \quad D = \{(x^s, x^s)\}_{s=1}^S, \qquad \Sigma = \Sigma_{xx}$$

$$E = \text{Tr}(\Sigma_{xx}) - \sum_{i \in J} \lambda_i \;\Rightarrow \text{PCA objective}$$

Baldi & Hornik:

$$X = \begin{bmatrix} \uparrow & & \uparrow \\ x^1 & \cdots & x^S \\ \downarrow & & \downarrow \end{bmatrix}$$

PCA: variance(true data) — var(proj. data)

Original source pages

Original scans for checking transcription and diagrams.

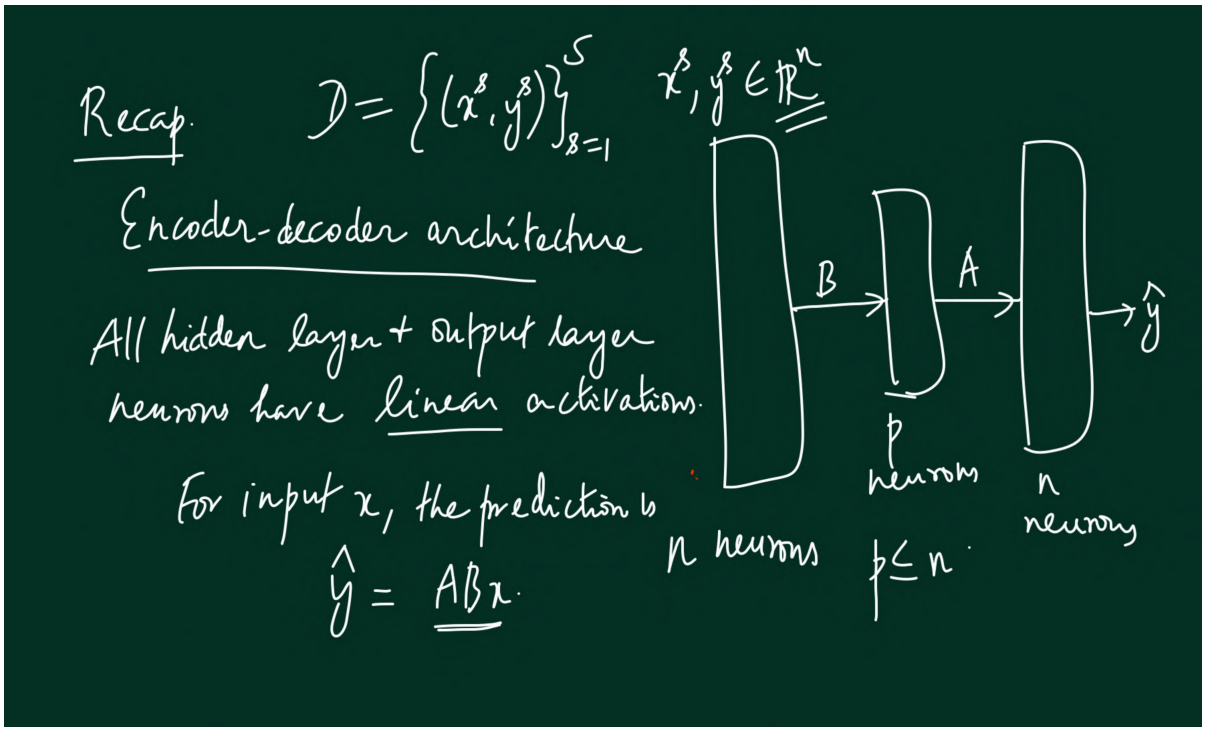


Figure 13: Original source page 1

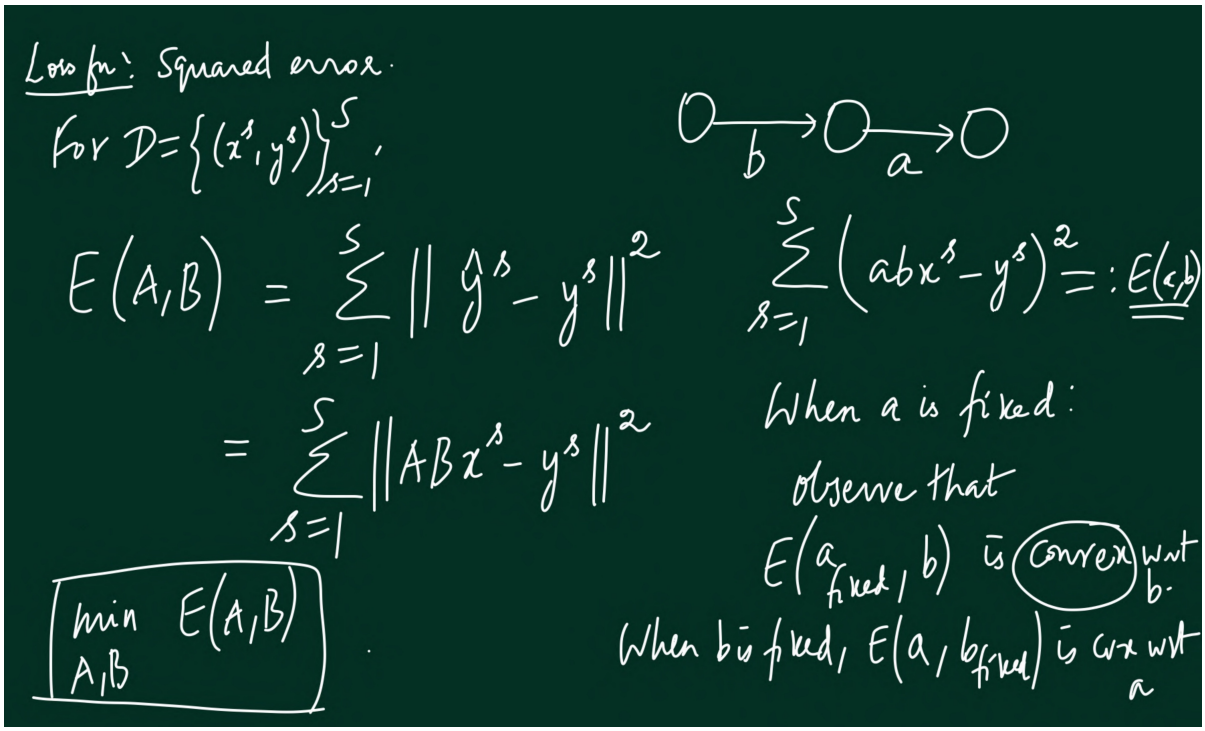


Figure 14: Original source page 2

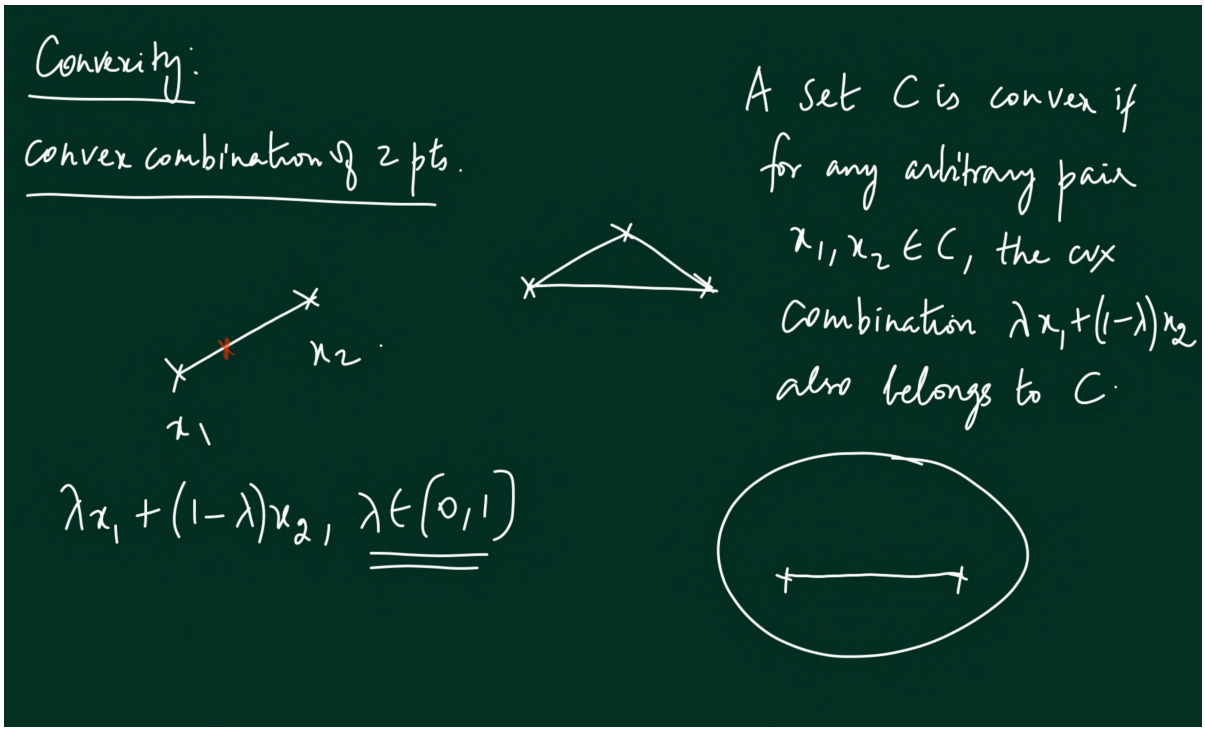


Figure 15: Original source page 3

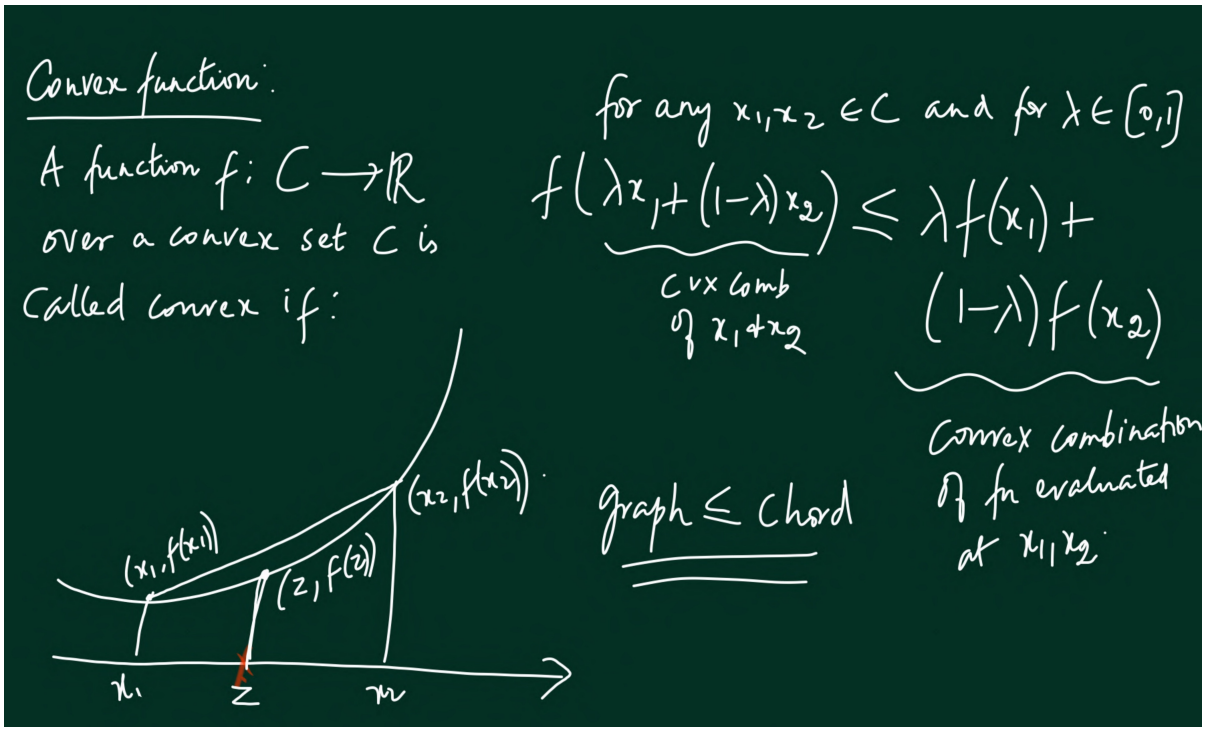


Figure 16: Original source page 4

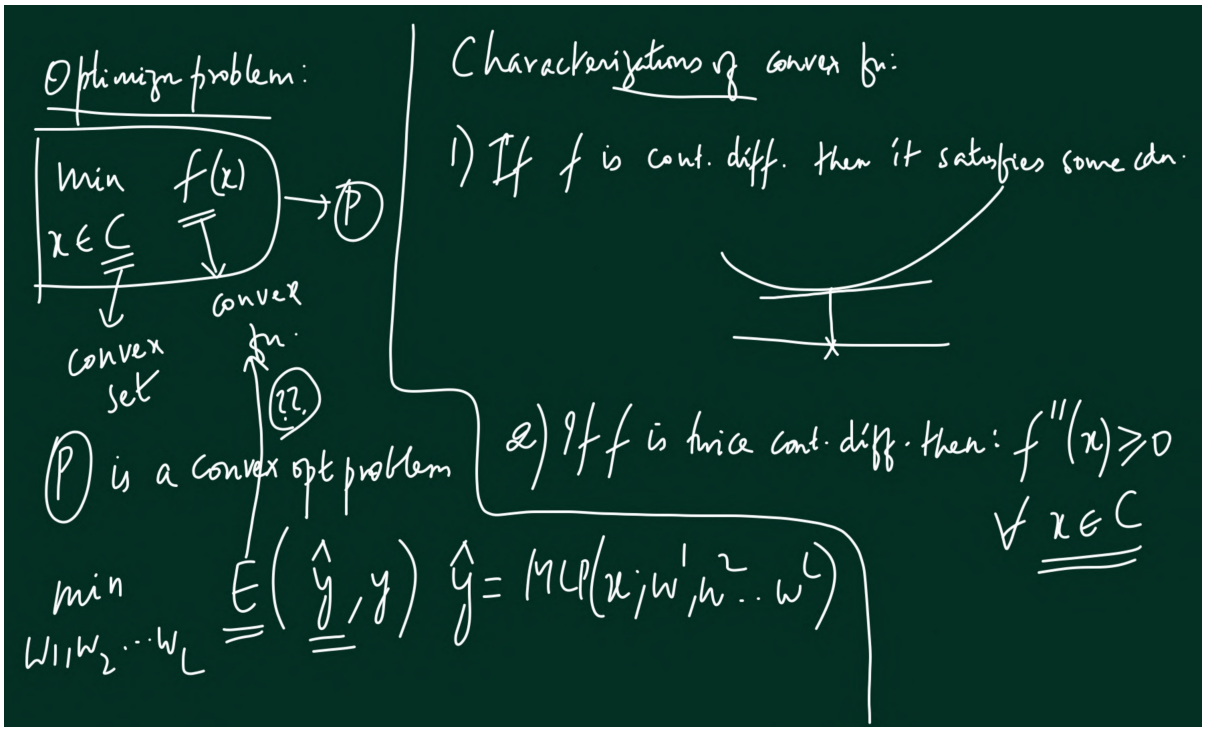


Figure 17: Original source page 5

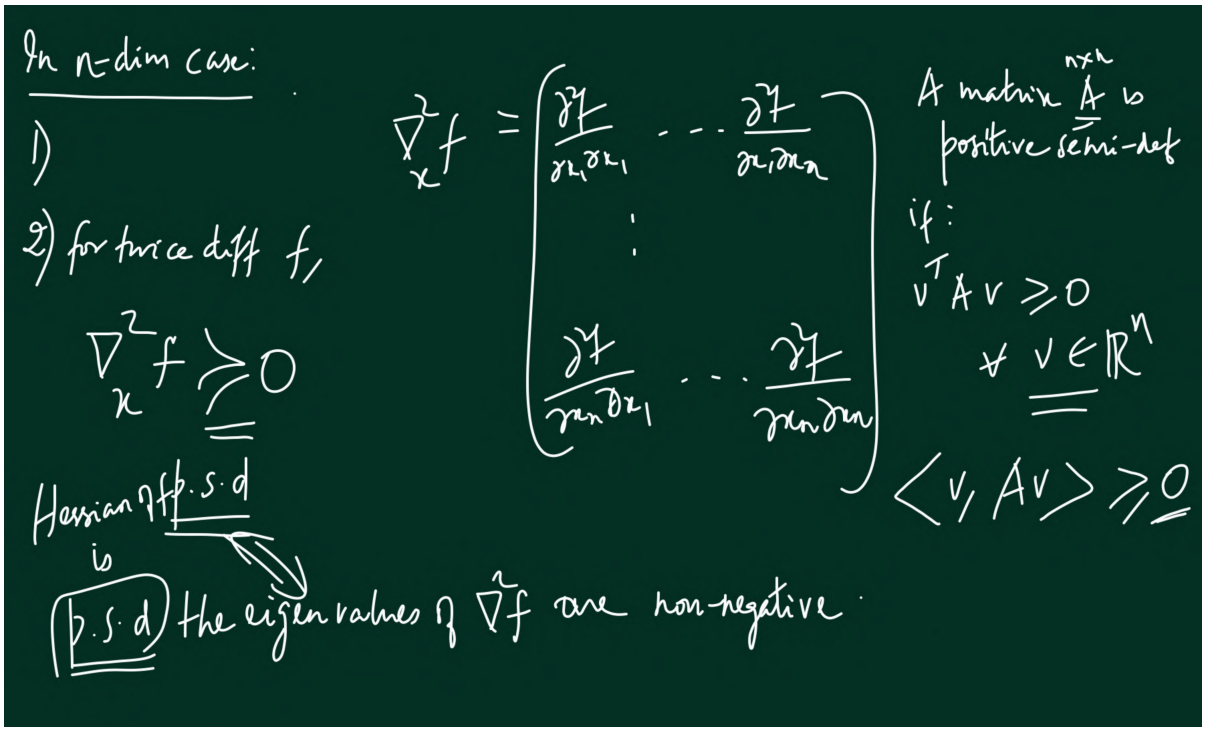


Figure 18: Original source page 6

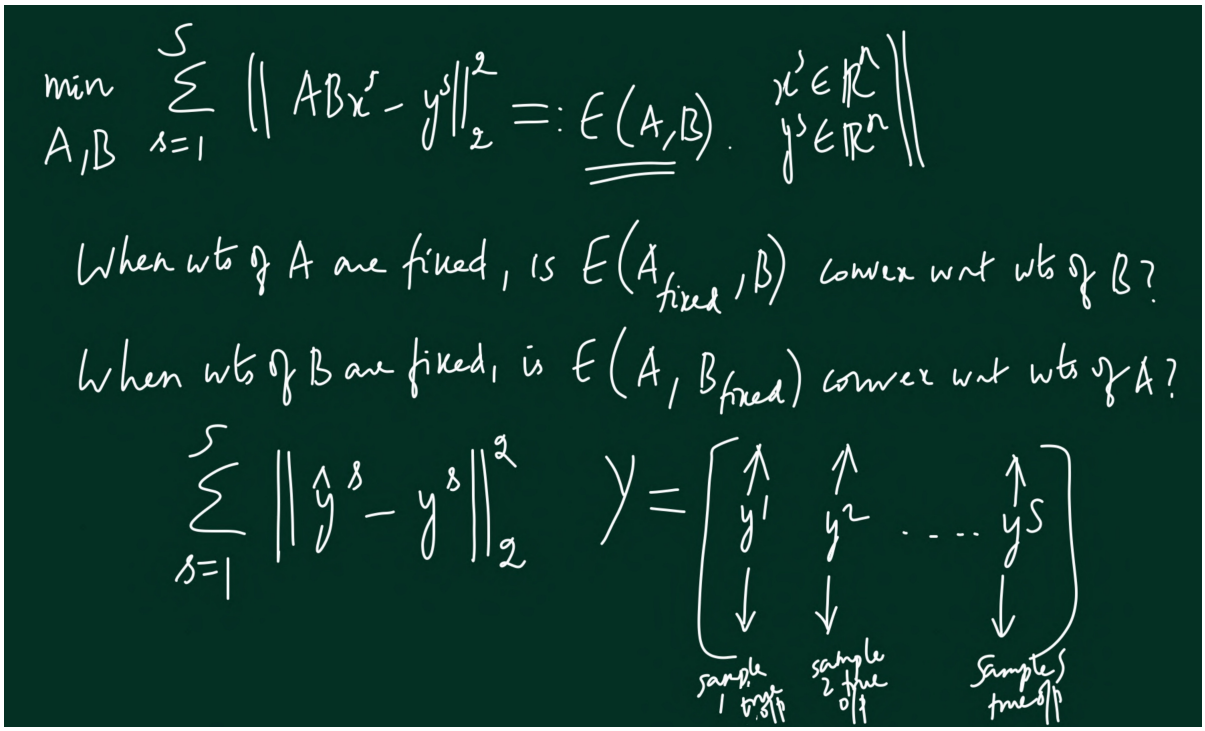


Figure 19: Original source page 7

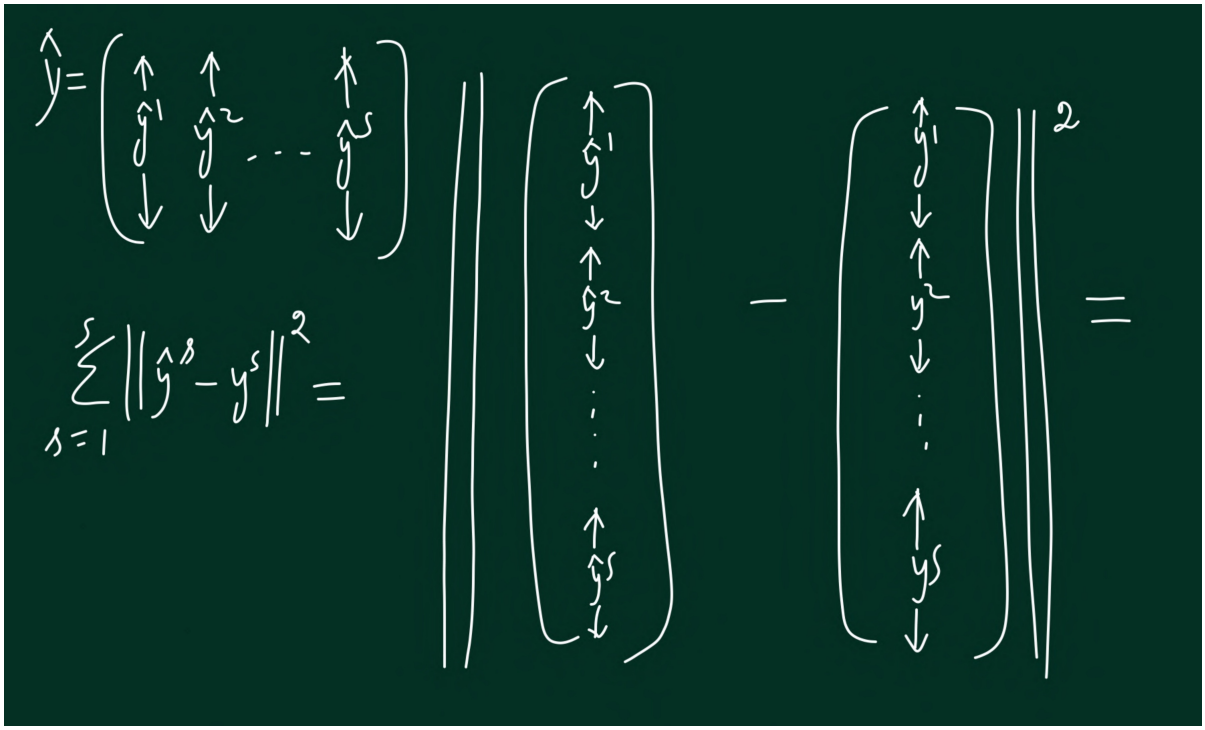


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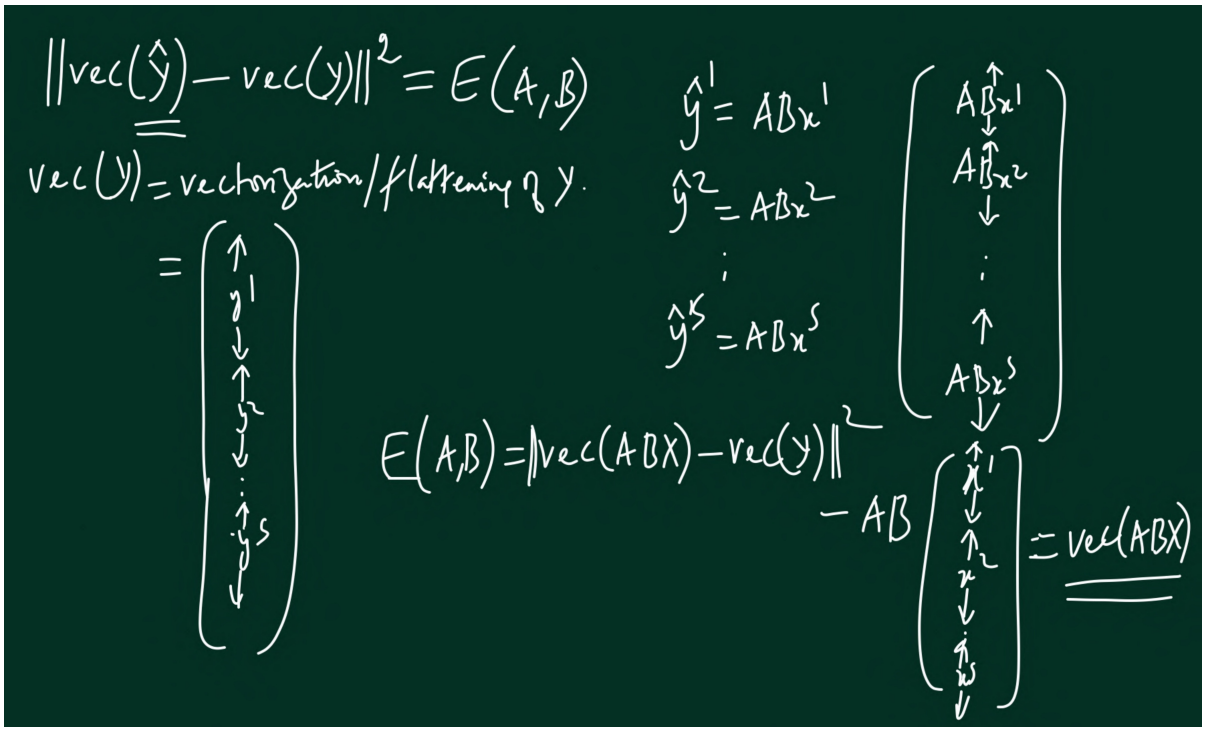


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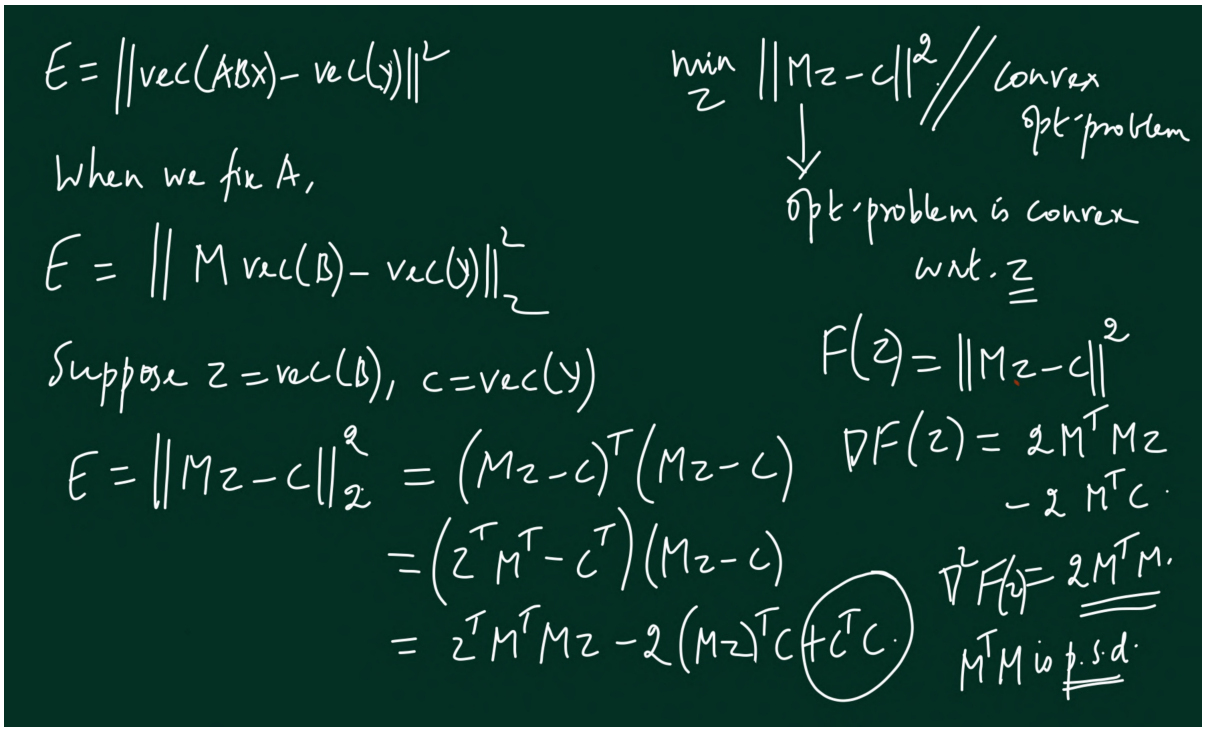


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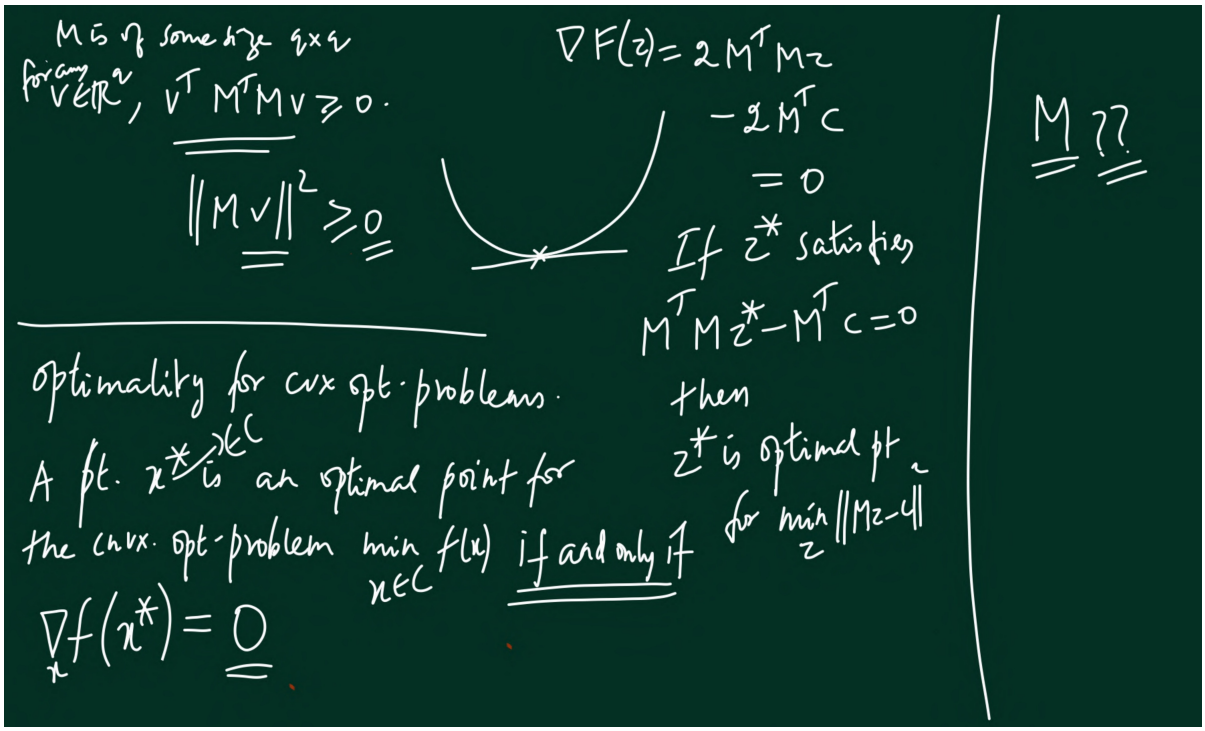


Figure 23: Original source page 11

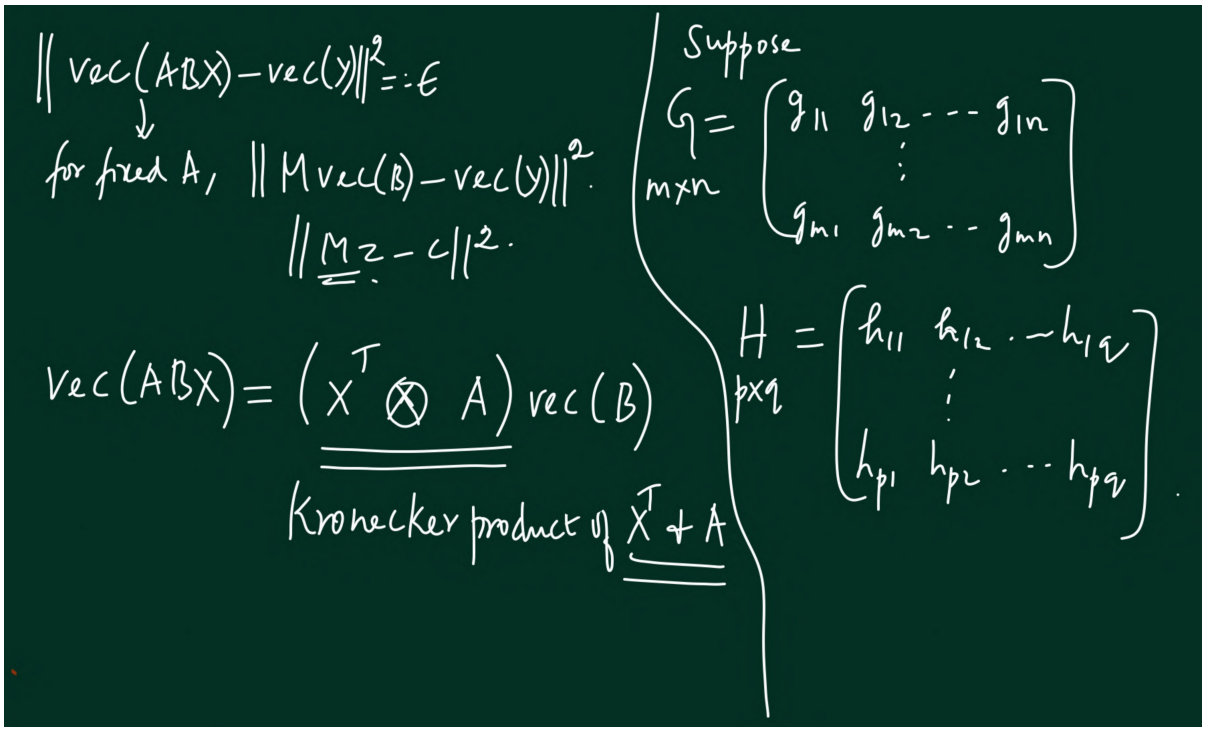


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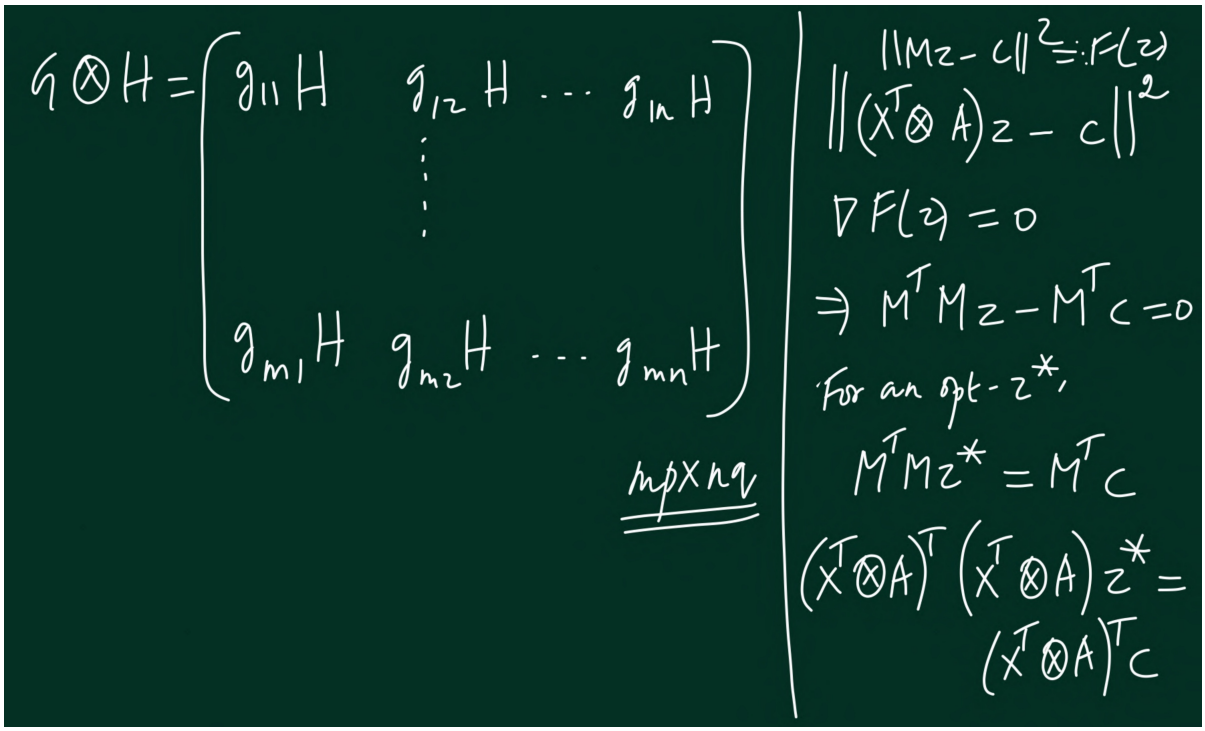


Figure 25: Original source page 13

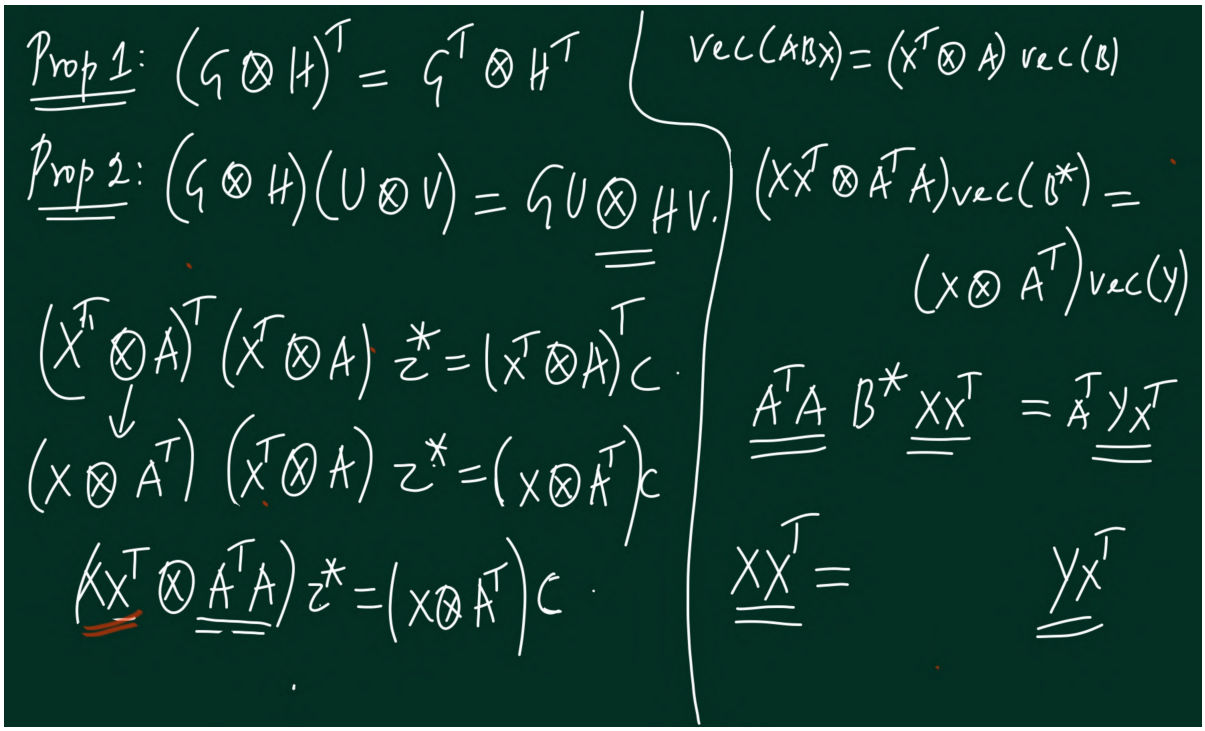


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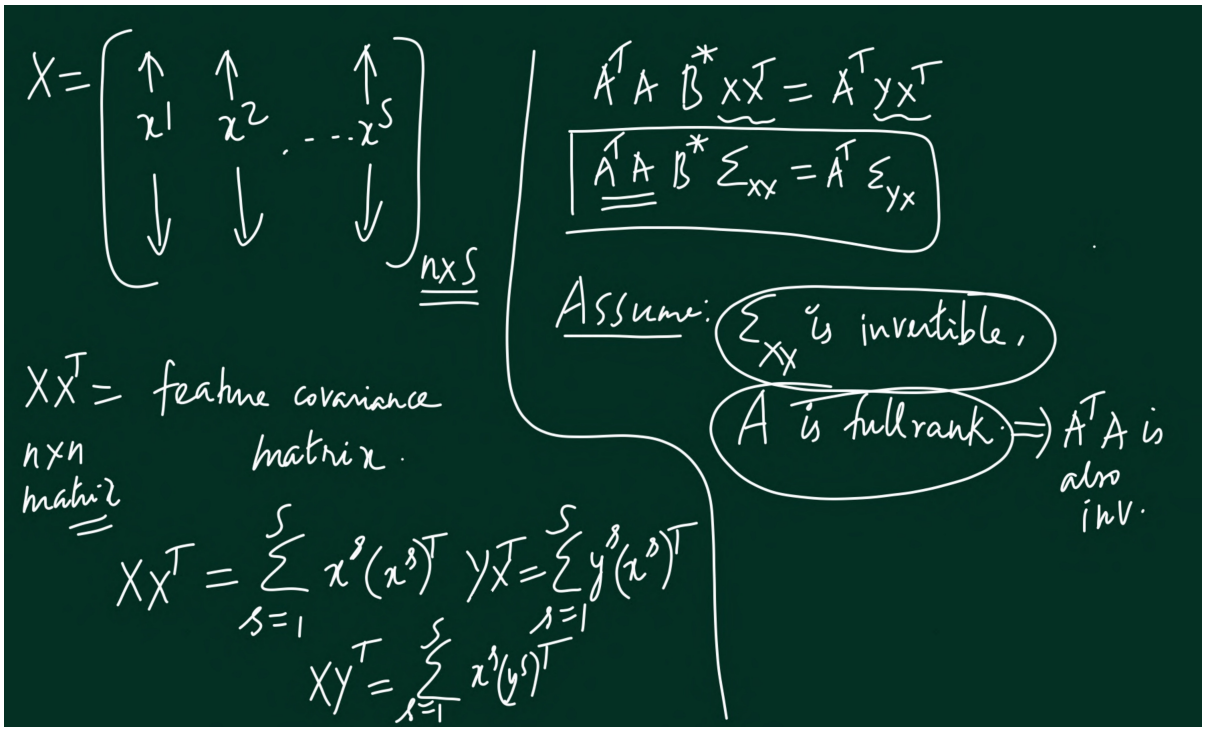


Figure 27: Original source page 15

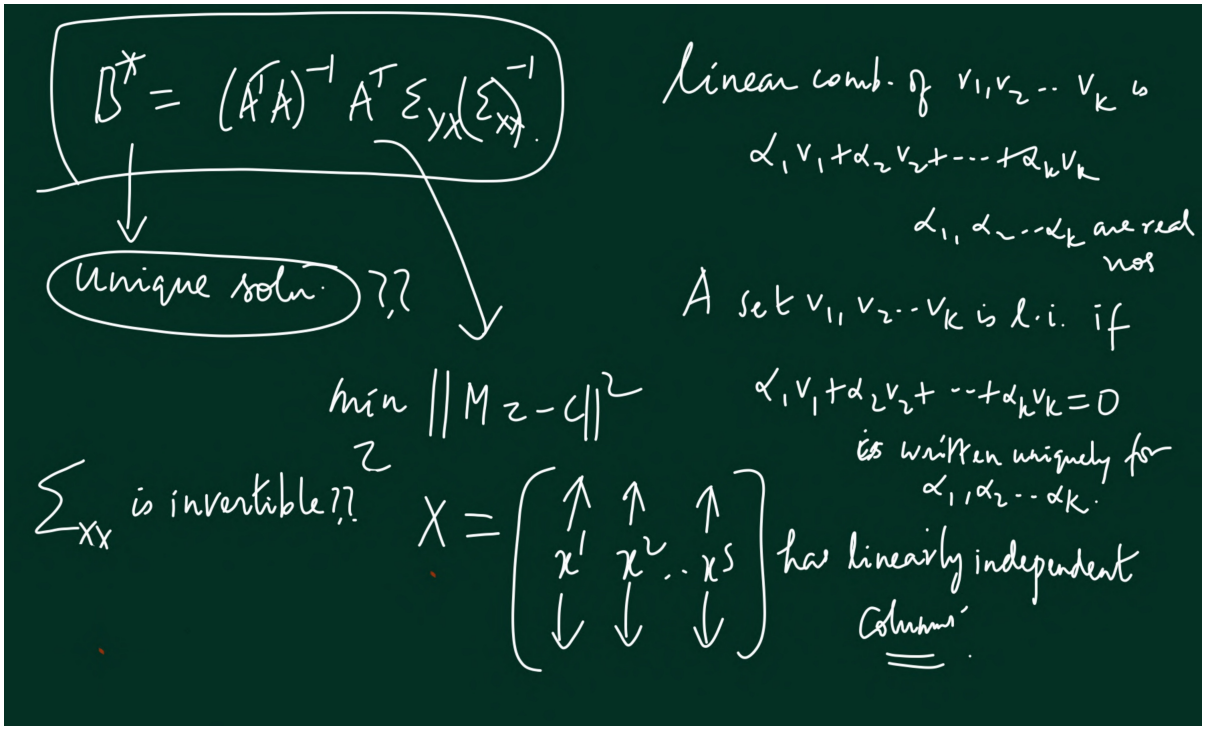


Figure 28: Original source page 16

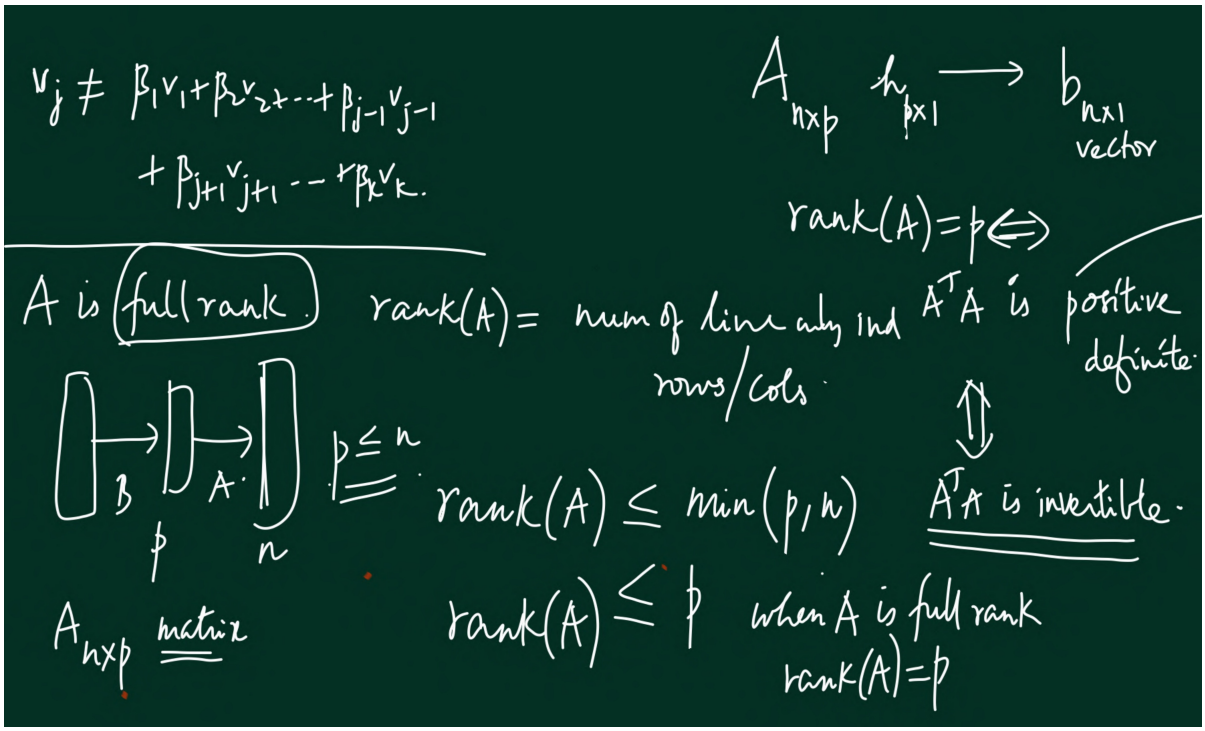


Figure 29: Original source page 17

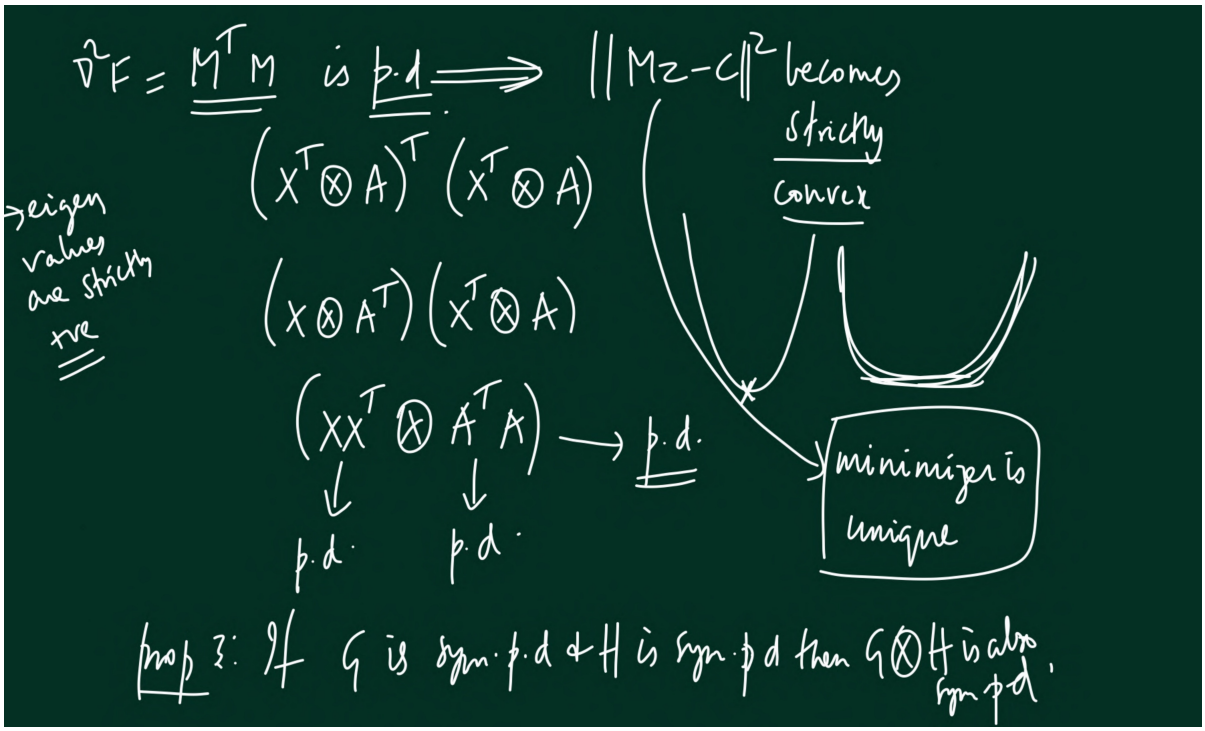


Figure 30: Original source page 18

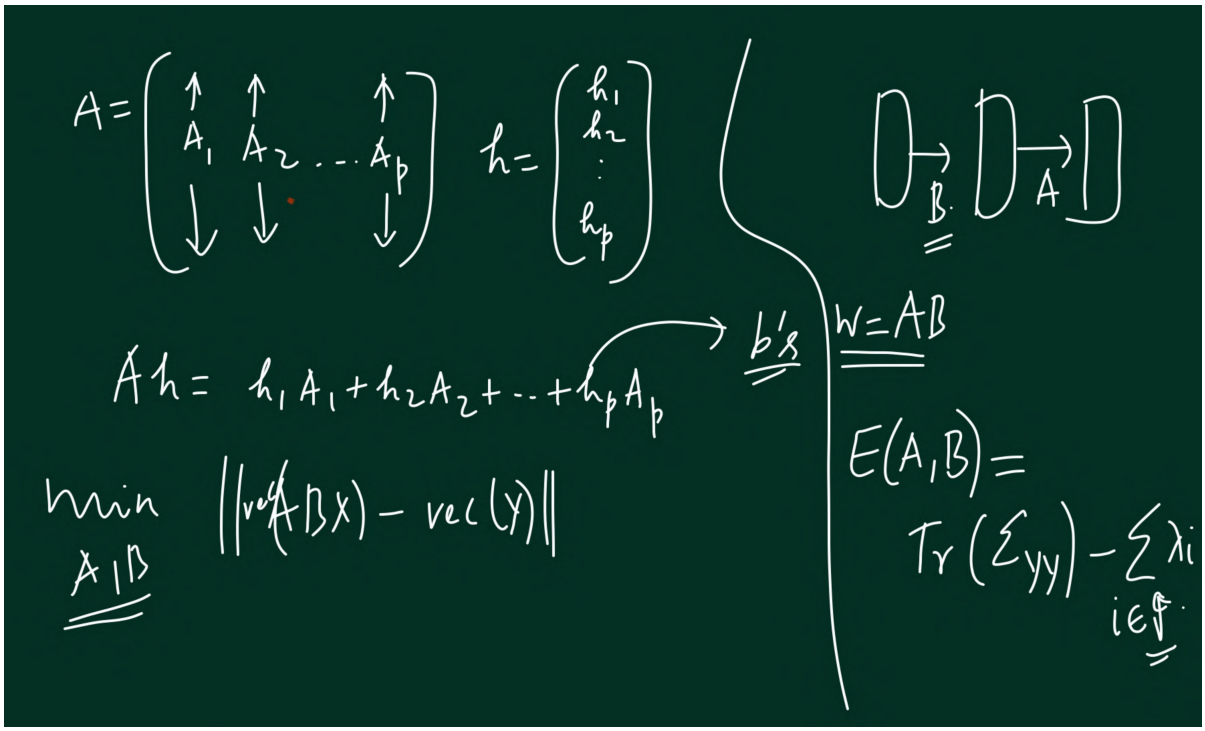


Figure 31: Original source page 19

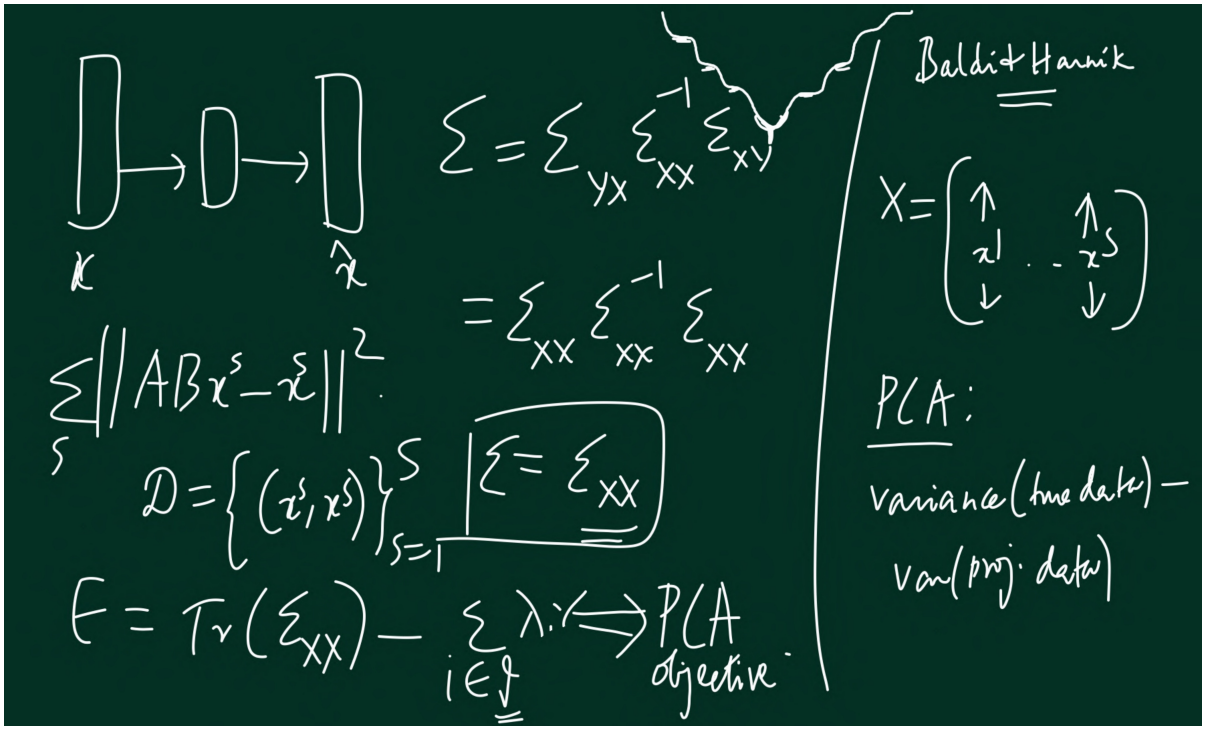


Figure 32: Original source page 20